

The Everest Bistro

Database Architecture & Schema Documentation (v4)

This document details the PostgreSQL 15+ / Supabase schema designed for The Everest Bistro. It outlines all 10 tables, 2 analytics views, and the 12 custom functions/triggers that enforce strict 3rd Normal Form (3NF), financial immutability, and state-machine transitions.

1. Configuration & Staff Tables

restaurant_settings

A constrained singleton table (one row only) holding global fiscal policies. Ensures that rate changes (e.g., VAT changes next quarter) do not rewrite historical financial records.

Column	Type	Description
id	BOOLEAN	Primary key, constrained to always be `true` to ensure a single row.
business_timezone	TEXT	Defaults to 'Asia/Kathmandu'. Used for accurate daily analytics aggregation.
default_tax_rate	NUMERIC(5,4)	Global tax rate (e.g., 0.1300 for 13% VAT).
default_service_rate	NUMERIC(5,4)	Global service charge rate (e.g., 0.1000 for 10%).
service_charge_on_exempt_items	BOOLEAN	Policy determining if service charge applies to tax-exempt goods.
prices_include_tax	BOOLEAN	Flag indicating if menu prices are tax-inclusive or exclusive.

staff_users

Manages role-based authentication for internal restaurant staff (Admin and Kitchen).

Column	Type	Description
	UUID	Primary key.

staff_id		
username	CITEXT	Case-insensitive unique login identifier.
password_hash	TEXT	Secure credential storage.
role	staff_role	Enum: 'ADMIN' or 'KITCHEN'. Strictly controls data access.

2. Table & Session Management

restaurant_tables

Represents physical dining tables. The status column is strictly derived from the table_sessions lifecycle to prevent orphaned states.

Column	Type	Description
table_id	UUID	Primary key.
qr_code_hash	UUID	Cryptographically secure, permanent static token embedded in the physical QR sticker.
status	table_status	Enum: 'EMPTY', 'OCCUPIED', 'CHECKOUT'. Read-only (Derived).

table_sessions

The core of the lifecycle. Tracks a seating event from start to finish. Ensures returning customers cannot access historical orders.

Column	Type	Description
session_id	UUID	Primary key. The dynamic token issued in the JWT upon QR scan.
table_id	UUID	Foreign key to the physical table.
status	session_status	Enum: 'ACTIVE', 'CHECKOUT', 'CLOSED', 'ABANDONED'. Defines the state machine.
device_count	SMALLINT	Telemetry tracking how many unique devices join a shared table session.

3. Menu Catalog System

categories

Hierarchical menu organization limited to 2 levels (Category → Sub-domain). Uses ON DELETE RESTRICT to prevent accidental mass deletion.

Column	Type	Description
parent_category_id	UUID	Self-referencing FK. NULL if top-level category.
name	VARCHAR	Category name (e.g., "Drinks").
archived_at	TIMESTAMPTZ	Soft-delete mechanism to preserve historical links.

menu_items

The master catalog of products. Contains current prices and current tax rules.

Column	Type	Description
category_id	UUID	Foreign key to categories.
price	NUMERIC(10,2)	Current active price.
is_out_of_stock	BOOLEAN	Temporary kitchen toggle to disable ordering.
is_tax_exempt	BOOLEAN	Defines if the item is excluded from VAT calculations.

4. Financial Ledger & Orders

orders

The master financial ledger for a single session. Stores the hierarchical calculation of subtotals, prorated discounts, and taxes.

Column	Type	Description
order_id	UUID	Primary key.

status	order_status	Enum: 'ACTIVE', 'COMPLETED', 'VOID'. Once completed, the order is locked.
subtotal	NUMERIC(10,2)	Gross sum of all non-cancelled items. (Auto-calculated).
taxable_subtotal	NUMERIC(10,2)	The portion of the subtotal subject to tax. (Auto-calculated).
discount_amount	NUMERIC(10,2)	Manual discount applied to the bill.
service_charge	NUMERIC(10,2)	Calculated automatically based on the snapshot service rate.
tax_amount	NUMERIC(10,2)	Calculated automatically based on the prorated taxable subtotal.
total_amount	NUMERIC(10,2)	The final, absolute amount owed by the customer.

order_tickets

Represents a "round" of ordering sent to the kitchen. Manages race conditions and network drops.

Column	Type	Description
ticket_number	INTEGER	Sequential order round (1, 2, 3) assigned safely via row locks.
idempotency_key	UUID	Client-generated UUID to block duplicate submissions from flaky connections.

order_items

Individual line items. Stores immutable snapshots of the price and tax rules at the exact moment of purchase.

Column	Type	Description
item_name_snapshot	VARCHAR	Preserved name of the item.
price_at_time	NUMERIC(10,2)	Preserved price of the item at purchase time.
is_taxable_at_time	BOOLEAN	Preserved tax status. Auto-populated by trigger if missing.
item_status	order_item_status	Enum: 'PENDING', 'DELIVERED', 'CANCELLED'.

The final settlement ledger. Strictly immutable. Represents cash/card/digital transactions resolving an order.

Column	Type	Description
order_id	UUID	Unique foreign key linking payment to the settled bill.
amount_tendered	NUMERIC(10,2)	Actual money handed over. Must be $\geq$ total_due.
change_given	NUMERIC(10,2)	Auto-calculated remainder returned to the customer.

5. Triggers & Stored Procedures (Business Logic)

The system enforces all operational and financial rules directly at the database layer via these functions:

- `enforce_derived_table_status()`: Prevents any manual modification to a physical table's status. It forces the system to transition states only through the `table_sessions` table.
- `enforce_session_transition()`: A strict state-machine guard. Ensures sessions can only flow legally: ACTIVE → CHECKOUT → CLOSED. Prevents closed sessions from reopening.
- `sync_table_status()`: An AFTER-trigger that automatically updates the physical table's color/status based on the current live session.
- `enforce_category_depth()`: Limits the menu hierarchy to exactly two levels to prevent UI layout breaks.
- `recalc_order_money()`: **Core Financial Engine**. Intercepts any changes to an order, splits the subtotal, prorates the discount across taxable/exempt lines, and safely calculates exact VAT and Service Charges.
- `lock_finalised_orders()` & `lock_items_on_finalised_orders()`: Freezes completed or voided bills. Prevents rogue API calls from altering historical financial data.
- `assign_ticket_number()`: Uses a `SELECT ... FOR UPDATE` lock to serialize concurrent kitchen orders from multiple devices, safely assigning sequential ticket numbers without race condition crashes.
- `snapshot_item_tax_status()`: Auto-populates a line item's tax status from the master catalog if the frontend API payload omits it.
- `recalc_order_subtotals()`: An AFTER-trigger on line items that recalculates both the gross and taxable subtotals and pushes them up to the parent order.
- `assert_payment_matches_order()`: Validates that a checkout payment exactly matches the outstanding order total before allowing the transaction to commit.
- `block_payment_mutation()`: Makes the payment ledger strictly append-only. Blocks UPDATE or DELETE commands.

6. Analytics Views

- `daily_sales`: Aggregates completed orders by the local business timezone. Splits revenue into gross, taxable, exempt, total discounts, service charges, and VAT.

- `daily_writeoffs`: Tracks all voided/walk-out orders to provide transparency into lost or written-off inventory.